Response to Reviewers — Third Round

Manuscript: "Lifetime Performance and Degradation of Floating Photovoltaic Systems Under Humid and Saline Conditions"

I thank the reviewer for the third assessment. This round implements the six priority revisions and the other suggestions. None of the reported results changes: the convergence analysis reuses exactly the 1,000 samples reported in the previous version as the first part of a 5,000-sample run.

Priority 1: Monte Carlo convergence (§21)

Comment 1 (convergence). *Show that 1,000 samples are sufficient, e.g. with 250–5,000 samples.*

Response. Done. A 5,000-sample Monte Carlo was run with the same seed. Its first 1,000 samples are identical to the reported analysis, so the statistics can be compared on nested subsets of 250, 500, 1,000, 2,000 and 5,000 samples, with bootstrap standard errors. From 1,000 to 5,000 samples, no median or 95% bound of the key outputs moves by more than 0.28 percentage point. The bootstrap standard errors at 1,000 samples are at most 0.30 percentage point, which is small compared with the interval widths of 6–12 points.

Changes: Section III-I (method); Section IV-H, Table XII and Fig. 13 (new); `uncertainty.convergence()` and a new unit test. The Monte Carlo now runs in parallel and is bit-identical to the serial run.

Priority 2: Assumptions versus empirical evidence (§6, §7, §19)

Comment 2 (model-evidence table). *Classify components into established models, literature-informed parameterizations and new assumptions.*

Response. Added as Table III, introduced at the start of Section III. It has three categories:

- A Established models used as published: Spencer, Haurwitz, Erbs, Hay–Davies, ASHRAE IAM, Kasten–Young, Gueymard, the First Solar spectral factor, Magnus, Faiman, PVWatts, the Arrhenius/Peck form, the ISO 9223 concepts and LCOE.
- B Literature-informed parameter values: heat-loss coefficients, Peck E_a and n, PID activation energy, NaCl deliquescence RH, ISO 9223 chloride classes, the dust coefficient, and reference rates calibrated to the magnitude of field surveys.
- C New modelling assumptions: the over-water microclimate, the surface-RH/encapsulant-lag formulation, the dew criterion and loss, salt deposition and cementation, the salt-assisted leakage function, the corrosion scaling, additive constant-rate degradation, the mitigation factors, and all deposition rates and costs.

The text notes that the contribution lies in the coupling and in category C, which is also where field calibration is most needed.

Comment 3 (PID submodel). *State that the PID submodel is a physically motivated scenario model, not an experimentally calibrated predictive model.*

Response. The sentence has been added to Section III-F in the reviewer's words. For readability, Eq. (13) now defines the shifted threshold $RH^*_{50} = RH_{50} - \Delta_s s$ and the width w explicitly.

Comment 4 (corrosion positioning). *Position the corrosion term carefully as a calibration-based representation.*

Response. Section III-F now states that Eq. (14) "borrows the ISO 9223 variables but not a fitted damage function". It adds that the linear time-of-wetness and square-root chloride scaling make it "a calibrated scenario representation of corrosion, not a validated corrosion-damage model".

Priority 3: Validation positioning (§3, §10, §11, §17)

Comment 5 (wording). *Use "baseline modelled cooling"; call Table III a "synthetic-weather consistency check"; keep the scenario framing; emphasize the validation protocol.*

Response. All are done:

- Section IV-A now reads "The baseline modelled cooling is 3.0 K in daytime ... This is a consequence of the assumed FPV heat-loss coefficients, not a measured cooling effect." The Discussion and conclusion use the same term.
- The weather table (now Table IV) is titled "Synthetic-Weather Consistency Check".
- The field-validation protocol is moved forward to Section V-C, directly after the decision-relevant drivers it is designed to measure. It is also listed as a contribution in Section I.

Priority 4: Constant-rate lifetime extrapolation (§12)

Comment 6 (constant rate). *Add a sentence immediately before Eq. (15) on equivalent constant-rate projections.*

Response. Added immediately before Eq. (15): "Consequently, the lifetime estimates should be interpreted as equivalent constant-rate projections rather than predictions of the exact annual degradation trajectory."

Priority 5: Baseline versus Monte Carlo labelling (§18)

Comment 7 (labelling). *Add the word "baseline" every time the $+2.1\%$ and -4.3% values appear.*

Response. Done. Every occurrence in the abstract, Sections IV-B, IV-E and IV-H, and the conclusion is now labelled "baseline". The Monte Carlo medians ($+1.5\%$ and -5.1%) are always reported with their 95% intervals.

Priority 6: Equation and typesetting audit (§20)

Comment 8 (typesetting). *Check Eqs. (9)–(14) and all subscripts, superscripts and units in the final PDF.*

Response. The compiled PDF was inspected at high resolution. Equations (9)–(14) render correctly with IEEEtran, so the compression the reviewer saw is most likely a PDF-to-text extraction artefact. Three equations were nevertheless restructured for readability:

- Eq. (13) is split over two lines with a named shifted threshold.
- Eq. (15) is written with a single sum $\sum(O_y - B_y)(1+r)^{-y}$, so that its equation number stays on the same line.
- The ambient-RH condition in the definition of TOW is stated explicitly.

The final PDF compiles without overfull boxes, and T'_a, T_m and the RH subscripts are used consistently.

Other comments

Comment 9 (§14, threshold wording). *Present $\Delta T_d^* \approx 2.7$ K as a model-predicted site-screening criterion, not a universal threshold.*

Response. Section IV-F now says it "is a threshold predicted by this model under the stated assumptions, not a universal physical constant; it can nevertheless serve as a site-screening criterion". The abstract, Section V-B and the conclusion say "in this model".

Comment 10 (§22, regional economics). *Phrase economic conclusions as "under the assumed Indian-market cost and financing conditions".*

Response. Done in Section IV-I (opening sentence and the justification sentence).

Comment 11 (§4, §13, preserve wording). *Retain the "model-based scenario predictions" and "scenario-based mitigation benefits" language.*

Response. Retained unchanged, and not strengthened.

Remaining limitation

Measured FPV and site-weather data are still unavailable, so the manuscript remains a physically informed scenario and uncertainty framework. Its evidential basis is now classified explicitly (Table III), and a field-validation protocol is proposed (Table XIV).